LaTeX Macro System Test Document

Test Suite

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1 Tier 1: Simple MathJax-Compatible Shortcuts

1.1 Number Systems

Basic number systems work: $\mathbf{Z}, \mathbf{Q}, \mathbf{R}, \mathbf{C}, \mathbf{F}, \mathbb{H}, \mathbb{N}$.

Field extensions: $\mathbf{F}_p, \mathbf{F}_{p^n}, \mathbf{F}_p, \mathbf{F}_{p^n}, \overline{\mathbf{Q}}, \mathbf{Z}_{\widehat{p}}$.

1.2 Calligraphic Letters

Categories: $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}$ and $\mathcal{T}\mathsf{op}, \mathcal{P}$.

1.3 Operators

Functors: $\mathrm{Hom}, \mathrm{Aut}, , \mathrm{Ext}, \mathrm{Tor}$.

Geometry: $\mathrm{Pic}, \mathrm{Spec}, \mathrm{Proj}, \mathrm{Div}$.

Galois theory: $\mathsf{Gal}, \mathrm{Frob}$.

2 Tier 2: Commands with Arguments

2.1 Delimiters

Absolute value: $|x|$, norm: $\|v\|$, inner product: $\langle u,\ v\rangle$.

Generators: $\langle a, b, c\rangle$, brackets: $\langle x, y\rangle$.

Floor and ceiling: $\lfloor 3.7 \rfloor = 3$, $\lceil 3.2 \rceil = 4$.

2.2 Algebraic Constructions

Polynomial ring: $[x, y, z]$, formal power series: t.

Quotients: ${}^{G}/_{H}$, localizations: $[S^{-1}]$.

2.3 Function Fields

Rational function field: (x), Laurent series: $(\!(t)\!)$.

2.4 Complexes

Chain complex: $C_\bullet$, cochain complex: $C^\bullet$.
Bicomplex: $K_{\bullet,\bullet}$.

3 Tier 3: Complex TeX Commands

3.1 Matrices

2x2 matrix: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
Column vector: $\begin{bmatrix} x \\ y \end{bmatrix}$

3.2 Stacking

Stacked symbols: $\substack{a \\ b \\ c}$

4 Tier 4: Package-Dependent Declarations

Using operators declared with `DeclareMathOperator`: $\operatorname{span} V$, $\operatorname{supp} f$, $\operatorname{Hilb} X$.

5 Category Theory Macros

5.1 Basic Categories

Categories: $\mathsf{Set}, \mathsf{Grp}, \mathsf{Ring}, \mathsf{Mod}, \mathsf{Top}, \mathsf{Vect}$.
Functors: $\operatorname{Hom}_{\mathcal{C}}(X, Y)$, $\operatorname{Aut}(G)$, (V).

5.2 Limits and Colimits

Limits: $\varprojlim F_n$, colimits: $\varinjlim G_n$.

5.3 Derived Categories

Derived functors: $\operatorname{Ext}^i(M, N)$, $\operatorname{Tor}_i(M, N)$.

6 Spectral Sequence Macros

Bordism spectra: $\mathrm{MO}, \mathrm{MSO}, \mathrm{MU}$.
K-theory: KO, KU.
Topological modular forms: $\mathrm{tmf}, \mathrm{TMF}$.

7 Mixed Examples

7.1 Algebraic Geometry

The Picard group $\mathrm{Pic}(X)$ classifies line bundles on X.

The spectrum $\operatorname{Spec} R$ of a ring R with structure sheaf $\mathcal{O}_X$.

Cohomology: $H^i(X, \mathcal{O}_X)$ and Chern classes $c_i \in \mathrm{CH}^i(X)$.

7.2 Number Theory

For a prime p, the p-adic integers are $\mathbf{Z}_{\widehat{p}}$ and p-adic numbers are $\mathbf{Q}_{\widehat{p}}$.

Galois group: $\mathsf{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ acts on roots.

7.3 Topology

Fundamental group: $\pi_1(X, x_0) \cong \mathbf{Z}$ for the circle.

Homology: $H_n(X; \mathbf{Z})$ and cohomology: $H^n(X; \mathbf{Z})$.

Homotopy groups: $\pi_n(S^n) \cong \mathbf{Z}$.

7.4 Representation Theory

Group ring: $\mathbf{Z}G$ where G is a group.

Character: $\chi : G \to \mathbf{C}^\times$.

Schur's lemma: $\mathrm{Hom}_G(V, W) = 0$ for inequivalent irreps.

7.5 Complex Analysis

Holomorphic functions on $\mathbf{C}$ with $|f(z)| \leq M$ are constant.

Residue: $\operatorname{Res}_{z=a} f(z)$.

7.6 Differential Geometry

Tangent bundle: TX, cotangent bundle: T^*X.

Exterior algebra: $\bigwedge V = \bigoplus_{k=0}^n \bigwedge^k V$.

8 Symbols and Relations

Embeddings: $X \hookrightarrow Y$, surjections: $X \twoheadrightarrow Y$.

Isomorphisms: $X \xrightarrow{\sim} Y$, homotopy equivalence: $X \simeq Y$.

Normal subgroup: $H \trianglelefteq G$.

Direct sum: $V \oplus W$, tensor product: $V \otimes W$.

9 Conclusion

This document exercises macros from:

- Tier 1: Simple shortcuts (number systems, operators)
- Tier 2: Commands with arguments (delimiters, constructions)
- Tier 3: Complex TeX (matrices, stacking)
- Tier 4: Package declarations (DeclareMathOperator)
- Domain files: Categories, spectral sequences

All macros loaded via `\input{koma-article}` which includes `dzg-unified.sty`.